



22AI503 – MACHINE LEARNING

UNIT 1 – INTRODUCTION

TOPIC 5: Regression - Linear Regression

5.1 Regression

Regression is a supervised machine learning technique used to predict a continuous numerical value based on one or more input variables.

It establishes a relationship between independent variables (features) and a dependent variable (target).

Definition

Regression is a statistical method used to model and analyze the relationship between variables, where the output is a continuous value.

Examples

- Predicting house prices
- Predicting stock prices
- Forecasting sales revenue
- Predicting student scores
- Estimating temperature

2. Need for Regression

Regression helps to:

- Predict future values
- Understand relationships among variables
- Identify significant factors affecting outcomes
- Support business decision-making
- Analyze trends and patterns

5.2 Types of Regression

1. Linear Regression

Relationship follows a straight line.

Example:

- Experience vs Salary
- Study Hours vs Marks

2. Multiple Linear Regression

Uses multiple input variables.

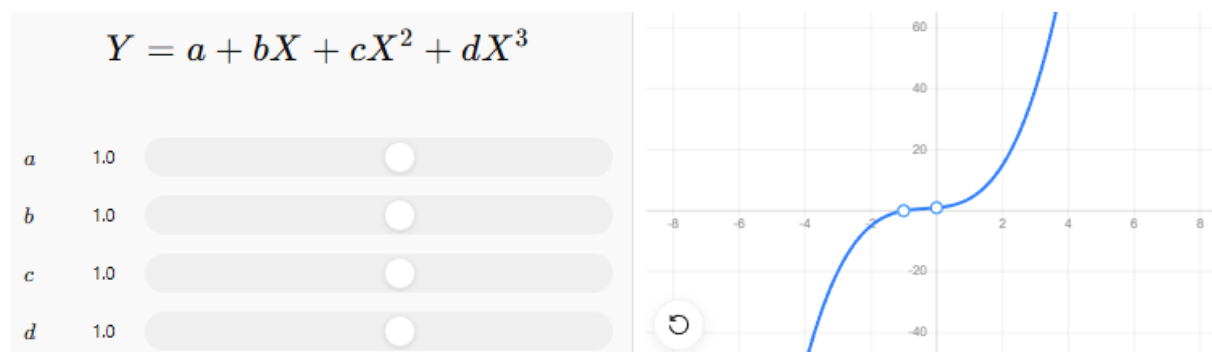
Example:

- House Price = $f(\text{Size, Location, Bedrooms})$

3. Polynomial Regression

Used when data follows a curved pattern.

Equation



Example:

- Population growth
- Vehicle fuel efficiency

4. Ridge Regression

Used when variables are highly correlated.

Advantages:

- Reduces overfitting
- Improves model stability

5. Lasso Regression

Adds regularization and performs feature selection.

Advantages:

- Removes less important features
- Simplifies models

6. Logistic Regression

Although called regression, it is used for classification.

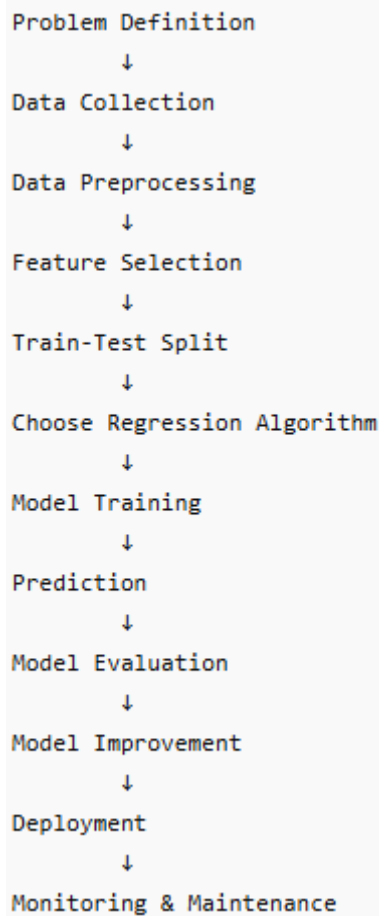
Example:

- Spam or Not Spam
- Disease or No Disease

Output:

- 0 or 1

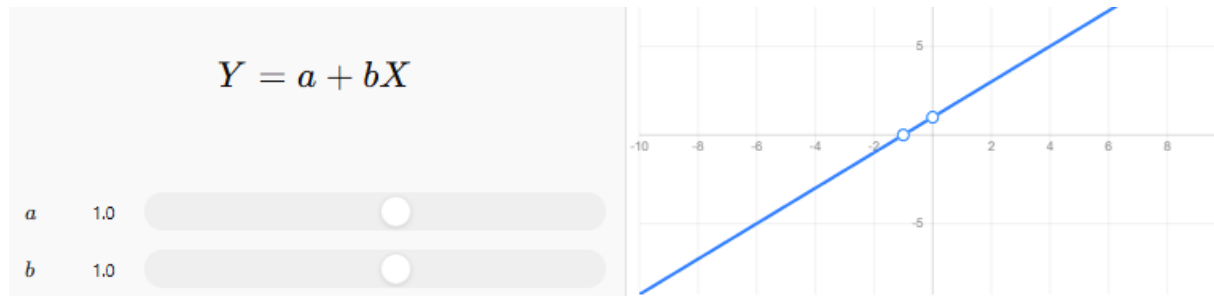
5.3 Workflow



5.4 Linear Regression in Machine Learning

Linear Regression is one of the simplest and most widely used supervised machine learning algorithms. It is used to predict a continuous numerical value by establishing a linear relationship between input variables and the output variable.

Definition: Linear Regression is a statistical method that models the relationship between a dependent variable and one or more independent variables using a straight-line equation.



Where:

- Y = Dependent Variable (Output)
- X = Independent Variable (Input)
- a = Intercept
- b = Slope (Regression Coefficient)

Meaning of Slope and Intercept

Intercept (a): The value of Y when X = 0.

Slope (b): The amount by which Y changes for every one-unit increase in X.

Example:

$$Y = 20 + 5X$$

- Intercept = 20
- Slope = 5

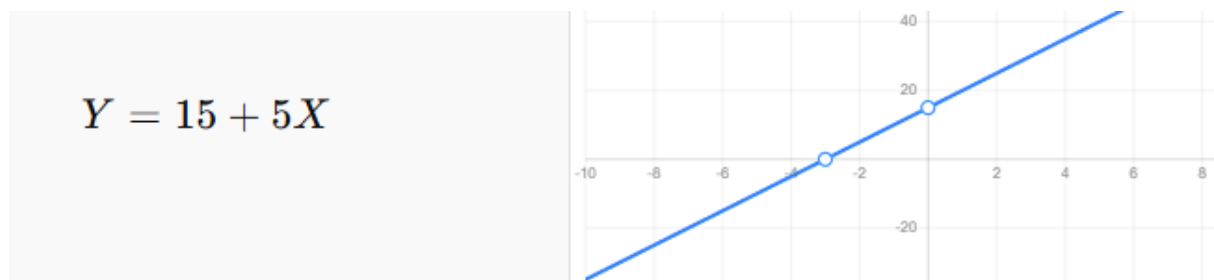
This means:

- Initial score = 20
- Every extra study hour increases score by 5 marks.

5.5 Solved Problems

Problem 1: Predict Student Score

The linear regression equation is:



where:

- X = Hours Studied



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- Y = Exam Score

Find the predicted score when a student studies for 8 hours.

Solution

Substitute $X = 8$

$$Y = 15 + 5(8)$$

$$Y = 15 + 40$$

$$Y = 55$$

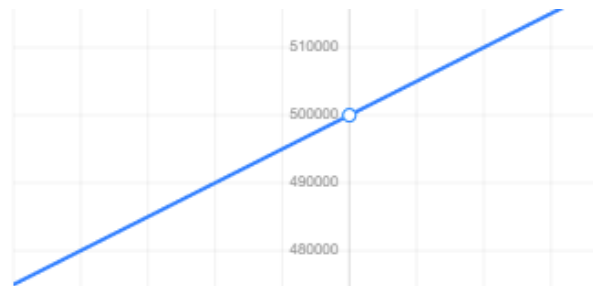
Answer

Predicted Score = 55 marks

Problem 2: House Price Prediction

A house price is modeled by:

$$Y = 500000 + 2500X$$



where:

- X = Area in square feet
- Y = House Price

Find the price of a house with an area of 1200 sq.ft.

Solution

$$Y = 500000 + 2500(1200)$$

$$Y = 500000 + 3000000$$

$$Y = 3500000$$

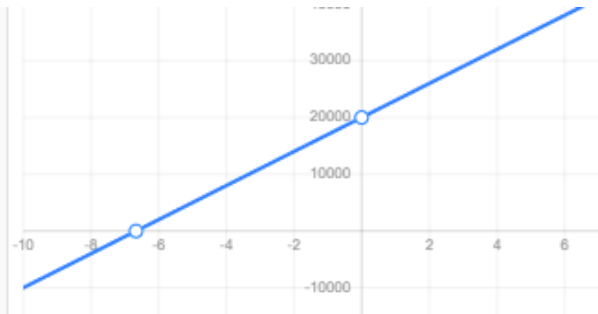
Answer

House Price = ₹35,00,000

Problem 3: Employee Salary Prediction

The salary model is:

$$Y = 20000 + 3000X$$



where:

- X = Years of Experience
- Y = Monthly Salary

Find the salary of an employee with 6 years of experience.

Solution

$$Y = 20000 + 3000(6)$$

$$Y = 20000 + 18000$$

$$Y = 38000$$

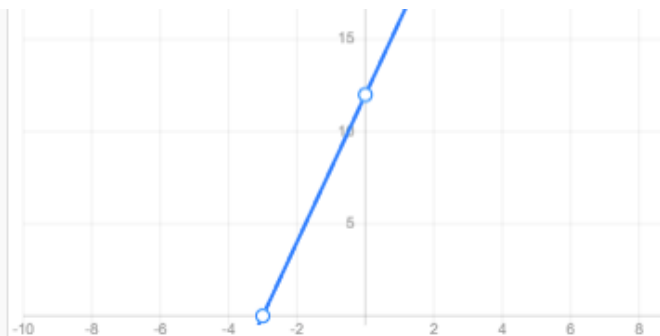
Answer

Salary = ₹38,000 per month

Problem 4: Find the Regression Coefficients

Given the regression equation:

$$Y = 12 + 4X$$



Identify:

1. Intercept
2. Slope

Solution

Comparing with:

$$Y = a + bX$$

- Intercept (a) = 12
- Slope (b) = 4

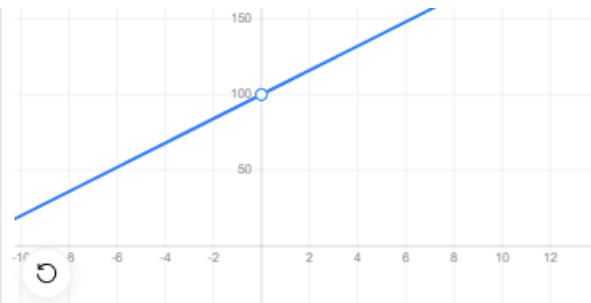
Answer

- Intercept = 12
- Slope = 4

Problem 5: Predict Sales Revenue

A company uses the regression model:

$$Y = 100 + 8X$$



where:

- X = Advertising Cost (₹1000)
- Y = Sales Revenue (₹1000)

Find sales revenue when advertising cost is ₹20,000.

Solution

$$X = 20$$

$$Y = 100 + 8(20)$$

$$Y = 100 + 160$$

$$Y = 260$$

Answer

Sales Revenue = ₹2,60,000

DISCUSSION QUESTIONS:

1. Why is Linear Regression considered one of the most fundamental supervised learning algorithms in machine learning?
2. What is the significance of the slope and intercept in a linear regression equation?
3. Under what circumstances is Linear Regression more suitable than Polynomial Regression?
4. How can Linear Regression be used in real-world applications such as education, finance, and real estate?
5. What are the limitations of Linear Regression when dealing with complex datasets?

**6. Problem 1: Predict Student Attendance Impact**

A linear regression model for student performance is:

$$Y=25+4X$$

where:

- X = Number of classes attended
- Y = Performance score

Find the predicted performance score when a student attends 15 classes.

7. Problem 2: Predict Monthly Electricity Consumption

The electricity consumption model is:

$$Y=120+8X$$

where:

- X = Number of appliances used
- Y = Monthly electricity consumption (kWh)

Calculate the consumption when 10 appliances are used.

8. Problem 3: Predict Crop Yield

A farmer uses the regression model:

$$Y=200+30X$$

where:

- X = Amount of fertilizer used (kg)
- Y = Crop yield (kg)

Find the crop yield when 8 kg of fertilizer is applied.

9. Problem 4: Find Slope and Intercept

Given the regression equation:

$$Y=18+7X$$

Identify:

1. Intercept
2. Slope

10. Problem 5: Predict Mobile Phone Sales

A company models mobile sales using:

$$Y=500+25X$$

where:

- X = Advertising budget (₹1000)
- Y = Number of phones sold

Find the expected sales when the advertising budget is ₹12,000.